

RESEARCH ARTICLE

Plant community development following reclamation of oil sands mines using four cover soil types in northern Alberta

Amalesh Dhar^{1,2} , Philip G. Comeau¹, M. Anne Naeth¹, Bradley D. Pinno¹, Robert Vassov³

Understanding the effects of reclamation treatments on plant community development is an important step in setting realistic indicators and targets for reclamation of upland oil sands sites to forest ecosystems. We examine trends in cover, richness, evenness, and community composition for four cover soil types (clay over overburden, clay over tailings sand, peat-mineral mix over overburden, and peat-mineral mix over tailings sand) and natural boreal forests over a 20 year period in the mineable oil sands region of northern Alberta, Canada. Tree, shrub, and nonvascular plant species cover showed similar increases over time for all reclamation treatments, with corresponding declines in forb and graminoid cover with time. These trends resemble those in the natural boreal forests of the region and the trajectory of community development for the reclamation treatments appears to follow typical early successional trends for boreal forests. Species richness and diversity of natural forest differed significantly from reclamation treatments. Nonmetric multidimensional scaling ordination and multi-response permutation procedure revealed that species composition was not affected by reclamation treatment but clearly differed from natural forest. Analysis of species co-occurrence indicated random plant community assembly following reclamation, in contrast to a higher proportion of nonrandom plant community assembly in natural forests. Thus, reclaimed plant communities appear to be unstructured through year 20 and assembly is still in progress on these reclaimed sites.

Key words: boreal forest, ordination, species coexistence, species composition, succession and assembly

Implications for Practice

- The temporal changes in understory plant communities reveal that species composition in the reclaimed ecosystems would be much different from those naturally disturbed ecosystems.
- Although the trajectory of plant community development shows the typical early successional progress of boreal forests, random plant community assembly suggests that the communities remain unstructured 20 years after reclamation.
- The significant temporal changes in reclaimed ecosystems suggest that studies of ecosystem establishment and development on reclaimed areas should be conducted over the long term as these monitoring data will be useful in developing adaptive management strategies for ecosystem recovery.

Introduction

Stand replacing disturbances (e.g. fire, mining, harvesting) reset succession and alter ecological communities and the ecosystem services they provide (Millennium Ecosystem Assessment 2005; Zhang et al. 2014). Successful restoration of anthropogenically disturbed sites requires a thorough understanding of the processes and patterns regulating ecosystem progression (Temperton & Hobbs 2004) and application of practices

that re-create landforms and plant community assemblages that support long-term successional development and ecosystem resilience (Macdonald et al. 2015a). In the Canadian boreal forest, fast growing shade intolerant tree species (e.g. aspen, paper birch, jack pine, etc.) and early successional understory vegetation (e.g. fireweed, red raspberry, bluejoint grass, etc.) colonize the disturbed sites and diversity reaches a peak within 30–40 years after disturbance (Lieffers et al. 1999; Hart & Chen 2006). However, successional patterns after anthropogenic disturbance and reclamation of oil sands mining sites are not well documented.

Given the small number of tree species in boreal forests, understory vegetation contributes substantially to biodiversity and ecosystem functioning (Hart & Chen 2006; Zhang et al. 2017). Consequently, establishing communities with a diverse understory plant community is an important ecological

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¹Department of Renewable Resources, University of Alberta, Edmonton, AB T6G 2H1, Canada

²Address correspondence to A. Dhar, email amalesh@ualberta.ca

³Canadian Natural Resources, Fort McMurray, AB T9H 4W1, Canada

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indicator of restoration success (Grant & Koch 2007; Macdonald et al. 2015b). While ecological restoration strategies should be modeled on dominant local natural disturbance and ecosystem recovery processes (Hobbs & Harris 2001), the outcomes of restoration efforts often depend on multiple factors including abiotic site conditions (e.g. aspect, soil water content, texture, soil physical and chemical properties, cover soil types), biotic factors (e.g. propagules, microbial associations, competition, dispersal, nonnative species), and time since restoration (Macdonald et al. 2015a; Stuble et al. 2017). Thus an understanding of how these influence development of plant communities following reclamation is potentially useful in the design of reclamation programs that are targeted toward restoring resilient plant communities that are similar to natural plant communities in the area.

The concepts of succession and community assembly can explain underlying processes of plant community development following disturbance when examining the resulting sequence of communities (seral stages; Young 2000). Both concepts address the temporal dynamics of ecosystems after disturbance, with succession focusing on the dynamics of a system following initial colonization and assembly addressing arrival, establishment, and persistence of species in the ecosystem through abiotic and biotic interactions (Hobbs et al. 2014). Plant species may influence community dynamics through inhibition, competition, and facilitation (Bruno et al. 2003). For example, in some instances prevailing species can strongly compete with colonizing species thereby inhibiting establishment of species that have similar resource requirements or that are functionally similar (Tilman & Pacala 1993; Rees et al. 2001). When vegetation is sparse existing species can allow establishment of new species in the system as there are less competitive and inhibitory effects of one species on another (Tilman & Pacala 1993; Fargione et al. 2004). Facilitation occurs when one species supports establishment of another species. For example the presence of *Populus tremuloides* (Michx.) facilitates conifer regeneration and establishment of understory species by reducing exposure, frost, and winter injury (Filipescu & Comeau 2011) and can also improve soil drainage, water storage, and nutrient availability (Comeau et al. 2005). These interspecific interactions may vary over time and influence plant community recovery after a disturbance in different ways at different times (Bruno et al. 2003). These interactions can be described using species co-occurrence pattern analysis where pairwise negative and positive correlations between species occurrences help in understanding the community dynamics (e.g. inhibition, competition, facilitation; Gotelli 2000; Veech 2013).

Oil sands mining in northeastern Alberta creates one of the world's largest and most intense industrial disturbances, and has removed nearly 953 km² of boreal forests (Alberta Environment and Parks 2017). Mining disturbance includes clearing vegetation, and removal of topsoil and overburden materials to expose the underlying McMurray Formation, which contains the bituminous sands (Alberta Environment and Water 2012). Regulations require that mining companies reclaim the mining sites to diverse, self-sustaining boreal forest communities similar to those in the surrounding region (Alberta Environmental

Protection 1998). Reclamation following mine closure thus involves constructing landforms and restoring native boreal forest communities.

Comparison of ecosystem development following reclamation of oil sands mine sites with natural ecosystems is commonly used as a basis for evaluating reclamation success as natural sites are thought to be biologically stable and representative of predisturbance conditions (Hobbs & Norton 1996; White & Walker 1997). Although some studies illustrate variability among and/or between reclamation treatments and natural forests (Quideau et al. 2013; Pinno & Errington 2015; Pinno & Hawkes 2015), researchers have rarely addressed the temporal changes in understory plant communities following different reclamation treatments on different site types relative to natural forest. However, this understanding is critical as use of data from naturally disturbed sites can help to identify whether the response of the restored ecosystem is caused by the reclamation activity or by unassisted recovery (White & Walker 1997; Ruiz-Jaen & Aide 2005). The primary objective of our study is to examine development of plant communities on different reclamation treatments over time relative to natural forests based on species diversity, compositional changes, and species co-occurrence patterns.

Methods

Site Description

The study sites were located near Fort McMurray in north-eastern Alberta, Canada (Fig. S1). The natural boreal mixed wood forests in this region are dominated by mixtures of *Populus tremuloides* and *Picea glauca* (Moench) Voss with components of *Pinus banksiana* L., *Populus balsamifera* L., *Betula papyrifera* Marshall., and *Abies balsamea* (L.) Mill. (Natural Regions Committee 2006). The common plant community associations for the natural forests are lichen-jack pine in xeric site and blueberry jack pine-aspen/aspen—white spruce/white spruce-jack pine, or low-bush cranberry aspen/white spruce/aspen—white spruce in the mesic sites (Beckingham & Archibald 1996). Upland soils of this region are typically Grey Luvisols, Brunisols, and Gleysols (Beckingham & Archibald 1996). The climate of the region is continental with long cold winters and short warm summers. Mean maximum and minimum temperatures are −17.4°C and 17.1°C, respectively, with 418 mm (rainfall 316.3 mm and snowfall 134 cm) mean annual precipitation (Government of Canada 2017).

Data Source and Plot Design

For this study, we used data from the Long Term Plot Network (LTPN) established by the Cumulative Environmental Management Association, a multi-stakeholder group responsible for developing guidance documents that provide recommendations and best practices for land reclamation in Alberta. Plot selection and establishment for the LTPN was initiated in 2000 and plots were added annually through 2009. Several sources of

Table 1. Characteristics of oil sands reclamation treatment sites and natural forest.

Site Type	Cover Soil Type	Time Since Reclamation or Stand Age				
		<5	6–10	11–15	>15	50+
Xeric	Clay over overburden (CO)	—	—	—	—	
	Clay over tailings sand (CT)	1	1	—	—	
	Peat-mineral mix over overburden (PMMO)	—	—	1	1	
	Peat-mineral mix over tailings sand (PMMT)	1	2	1	1	
	Natural forest					21
Mesic	Clay over overburden	—	1	1	1	
	Clay over tailings sand	1	1	1	1	
	Peat-mineral mix over overburden (PMMO)	4	8	7	4	
	Peat-mineral mix over tailings sand (PMMT)	3	4	4	2	
	Natural forest					16
Total plot		10	17	15	10	37

information were used to select potential areas for establishment of a permanent plot. These included air photos of the region, regional soil survey reports, and available documentation of the distribution of natural forest stands, and reclaimed lands managed by the companies in Alberta oil sand extraction region. Once a suitable natural stand or reclaimed polygon was identified, a permanent sample plot location was randomly selected. Wherever possible the plot was positioned to maintain a minimum of 50 m between the plot boundaries and any adjacent clearings.

The LTPN data used in this study consist of 89 permanent plots located on xeric (dry) and mesic (moist) reclaimed and natural forest sites of which 52 are from oil sands reclamation sites with a variety of reclamation treatments and 37 are from natural forest sites (Hawkes et al. 2012). The natural forest sites were used as a basis for evaluating plant community development following reclamation (Table 1). The age of reclamation sites was 2–24 years and natural forest sites were more than 50 years of age. Although the age of natural forest sites is not ideal for comparison with reclaimed sites, there are currently no data available for younger stands. Plots were initially established to study the range of reclamation practices and site types common in the region. Measurements began in 2000 and continued through 2009. Some reclaimed plots were assessed on a 5-year interval resulting in a total of 81 measurements for 52 plots.

Reclamation Treatment and Data Collection

Reclaimed plots had four distinct reclamation treatments: clay soil over finer textured (45–50% sand) overburden (CO), clay soil over coarser textured (90–95% sand) tailings sand (CT), peat-mineral mix soil over finer textured overburden (PMMO), and peat-mineral mix soil over coarser textured tailings sand (PMMT; see Table 1). Overburden is a mixture of soil and organic material that was salvaged during mine advance; tailings sand is a mixture of sand, fine silts, clay, residual bitumen, lighter hydrocarbons, inorganic salts, and water-soluble organic compounds that is a by-product of the bitumen extraction process. The cover soil placement depth of peat-mineral mix and clay soil were 20 and 30 cm, respectively. All reclaimed and

natural forest plots were grouped into mesic and xeric categories of ecosites, based on topography, parent material, soil drainage, water table depth, and texture (PSECI 2010). After cover soil placement, older sites were planted primarily with nursery grown conifers (e.g. white spruce and jack pine) and newer sites also included planting of trembling aspen. Sites were fertilized (31.5-16-5 N-P-K at 250–300 kg/ha) and seeded with cover crops (e.g. barley) following typical oil sands reclamation operations (Hawkes et al. 2012).

The understory plant community was characterized in ten 0.4 × 1.0-m subplots systematically located inside each of 89 main plots (10 × 40 m; Fig. S2). Plant species cover was determined for each species of shrub, forb, and graminoid (grasses, sedges, rushes) in the subplot. Total ground cover was estimated for each layer and for all vegetation combined. Tree layer cover was estimated in four 10 × 10-m subplots in each main plot. The total number and cover of nonnative species were estimated. Nomenclature of vascular plants follows Moss (1983) as does the description of life strategy and growth form (trees, shrubs, forbs, graminoids) and designation of a species as native and nonnative to Alberta. No soil related data were obtained.

Statistical Analyses

Vegetation cover percent and diversity indices, including species richness (S), Shannon-Weiner index (H'), and Shannon evenness (E), were used to compare treatments for shrub, forb, and graminoid groups. The tree layer was not included in calculation of diversity indices due to small numbers of tree species in the dataset. Plot averages based on data from the 10 subplots (sampling units) were treated as the experimental unit in the data analysis.

When reclaimed plots were grouped into four age classes (<5, 6–10, >10–15, >15 years) for each site type for analysis, all treatments in xeric sites and CO and CT treatment in mesic site had small numbers of plots which limit statistical power. Therefore, mesic PMMO, mesic PMMT, and natural forest were used for final analysis. Results from the CO and CT treatments are discussed in the manuscript to illustrate observed trends relative to mesic PMMO, mesic PMMT, and natural

forest. Although some plots had repeated measurements, only the last measurement of a plot was included in the analysis to remove issues relating to plot independence. In total, two sets of mixed effects models were developed as the natural forest had only one age group whereas reclamation sites have four age groups. In the first model (Model I) only reclamation treatments (not including natural forest) and reclamation age group were treated as fixed effects; in the second model (Model II) reclamation treatments including natural forest were treated as fixed effects (reclamation age group ignored). In both models, plots were treated as a random effect. During the analysis we were unable to include interactions (time $\times$ treatment) due to the limited number and representation of observations for some categories (Table 1). Analyses were conducted using the function *lme* from the package *nlme* in the R statistics system version 3.2.5 (R Core Team 2016). We also used spaghetti plots to examine temporal trends in plant community development for mesic PMMO ($n = 11$) and PMMT ($n = 6$) treatment plots that were repeatedly measured. The spaghetti plots were created using the *CorrMixed* 0.1-13 within the linear mixed-effects model function (Van der Elst et al. 2016). Post hoc analysis was conducted using the function *glht* from the package *multcomp* 1.3-2 if significant mean differences were found. Normality and homogeneity of variances were tested by examining residuals versus fitted plots and normal q - q plots of the models. Square root transformation was applied when needed.

An unconstrained nonmetric multidimensional scaling (NMDS) ordination was used to illustrate variation in plant community composition among reclamation treatment and the natural forests. To support the descriptive ordination and to quantify multivariate differences in species composition a multi-response permutation procedure (MRPP) was used. Ordination and MRPP were conducted using the package *vegan* 2.4-2 (Oksanen et al. 2017) in R. To determine species that significantly contributed to the ordination axes an equilibrium circle was drawn on the principal component analysis (PCA) platform using Biodiversity R version 2.8-3 (Kindt & Coe 2005). The circle has a radius equal to $\sqrt{(d/p)}$, where d equals the number of dimensions of the reduced space considered (usually $d = 2$) and p equals the total number of descriptors (hence of principal components) in the analysis.

The goal of investigating species co-occurrence patterns was to provide a general idea of plant community assembly and not to identify causes of or processes related to species associations on oil sands reclamation sites. The species co-occurrence patterns were analyzed using the probabilistic model of species co-occurrence (Veech 2013) which allows analytical (without randomization or simulation) determination of the probability that two selected species co-occur at a frequency either less than or greater than the observed frequency of co-occurrence. These probabilities can be obtained analytically under the condition where a species probability of occurrence at each site is equal to its observed frequency among all sites. When the observed frequency of co-occurrence is significantly large and greater than expected a positive association is indicated; if co-occurrence is significantly small and less than expected a negative association is indicated; or if observed and expected co-occurrence is not

significantly different and approximately equal to expected a random association is indicated (Veech 2013). We conducted analyses to determine the pairwise association between tree and understory species. Analysis of species co-occurrence patterns was conducted using the R package *co-occur* 1.3 (Griffith et al. 2016).

Results

Cover and Diversity

Mean percent cover of trees, shrubs, and nonvascular species (Fig. 1A, 1B, & 1E) showed similar increases over time, whereas forb and graminoid cover declined with time (Fig. 1C & 1D). A similar pattern of change was observed when spaghetti plots were created to examine the temporal trends in plant community development for the plots that had two time periods measured in PMMO and PMMT treatments (Fig. S3). Cover of the shrub, forb, and graminoid layers did not differ significantly between PMMO and PMMT treatments but did differ with time since reclamation for forb ($p < 0.001$) and graminoids ($p < 0.001$; Model I). When the natural forest condition was compared to the reclamation sites (Model II), a significant difference was observed in shrub ($p < 0.001$) and graminoid ($p = 0.013$) layers, whereas no differences were detected for the forbs layer (Table S1). Post hoc comparisons showed that shrub and graminoid cover in mesic PMMO and PMMT treatment differed significantly ($p < 0.001$ and $p < 0.001$) from the mesic natural forest.

Both native and nonnative species cover declined on PMMO and PMMT treatments over time (Fig. 2A & 2B). Greatest cover of nonnative species was found in PMMT at age groups <5 years (55%), followed by CT (44%) and PMMO (26%) at age group 6–10 years. After 16 years cover of nonnative species was 19% in CT (clay tailings), 9% in PMMT (peat-mineral mix tailings), 8% in PMMO (peat-mineral mix overburden), and 5% in CO (clay overburden) treatments (Fig. 2B). A similar declining trend was also observed when spaghetti plots were created for the repeated measured mesic PMMO and PMMT plots (Fig. 3G & 3H). Overall native species cover in all reclamation treatments was significantly lower than in natural forest (Fig. 3B).

Species richness and diversity (Shannon diversity and evenness) for forbs and graminoids did not differ significantly between reclamation treatments (mesic PMMO and PMMT) but differed among the reclamation age groups (forb Shannon [$p = 0.002$] and richness [0.003]; graminoids Shannon [$p = 0.003$], richness $p < 0.001$]; Model I; Table S1). Richness and diversity did not change much over time but showed a slight declining trend (Fig. 3). Shrub species richness increased slightly over time (PMTT and PMMO) but remained substantially below natural forest values (Fig. 3B). Forb richness also declined with time (Fig. 3C) and after 16 years, forb richness was significantly ($p = 0.004$) lower in reclaimed areas than in the natural forest. Graminoid richness in natural forest was similar to PMMO and significantly lower than PMMT ($p = 0.025$; Fig. 3D). When reclamation treatments were compared with natural forest for diversity, shrub (both Shannon and Shannon

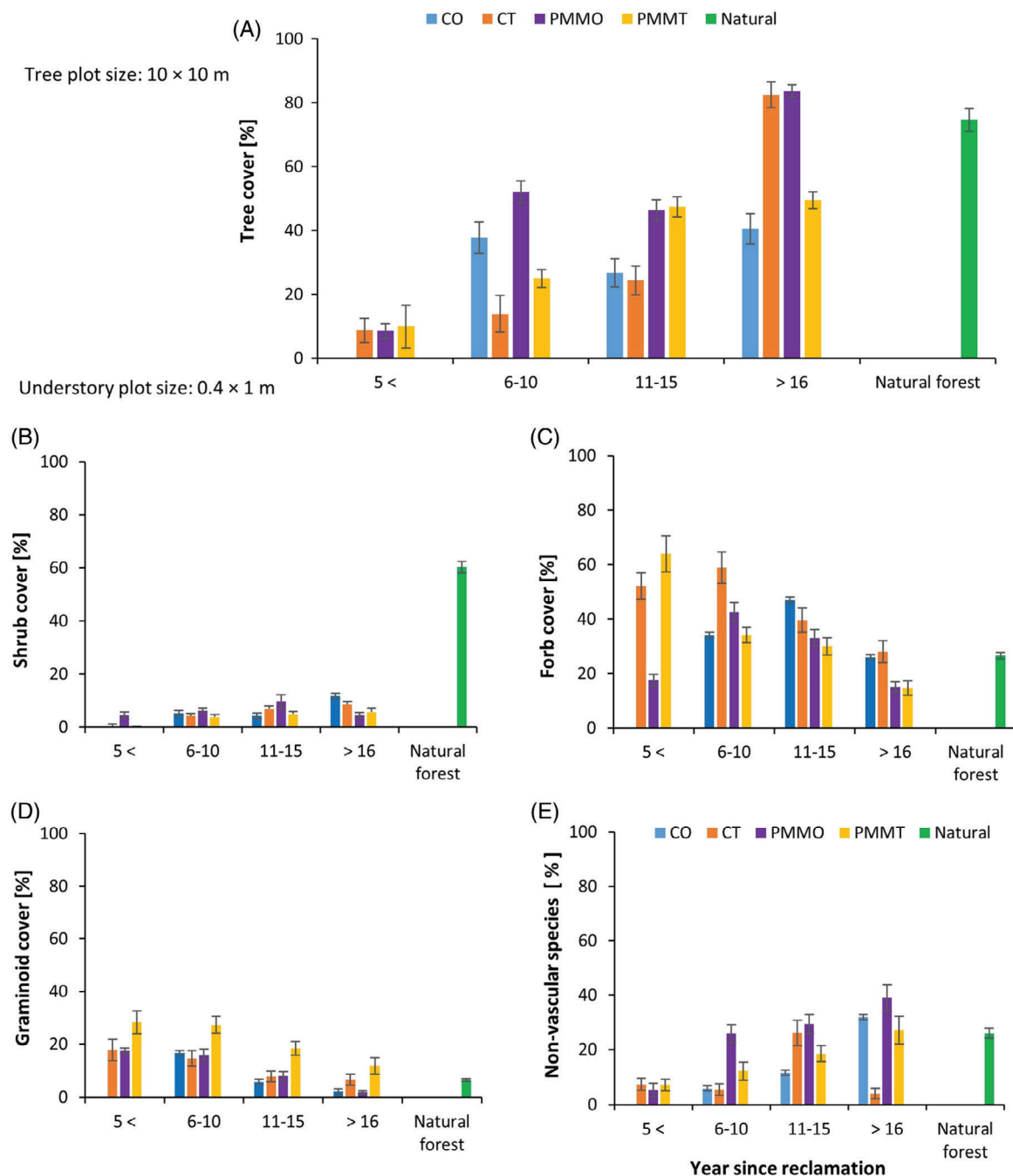


Figure 1. Mean ($\pm$ SE) percent cover by soil treatments relative to natural forest and age for tree (A), shrub (B), forb (C), graminoid (D), and nonvascular (moss and lichen; E) species in oil sands reclamation sites. CO, clay over overburden; CT, clay over tailings sand; natural, natural forest; PMMO, peat-mineral mix over overburden; PMMT, peat-mineral mix over tailings sand.

evenness $p < 0.001$), forb (Shanno $p < 0.001$), and graminoid (Shannon $p < 0.001$ and Shannon evenness $p = 0.003$) layers differed significantly except for Shannon evenness for forb layer (Table S1). Post hoc comparisons also showed that natural forest differed significantly from PMMO and PMMT for diversity and richness in most instances. Total species richness and diversity for nonnative species (Fig. 2D, 2F, & 2H) were significantly lower than native species (Fig. 2C, 2E, & 2G) in

most of the reclamation treatments and years since reclamation. The mean nonnative species richness per plot was below 0.02 in the natural forest.

Species Composition

In total, 219 understory plant species and eight tree species were found in the reclamation treatments and natural forest, which

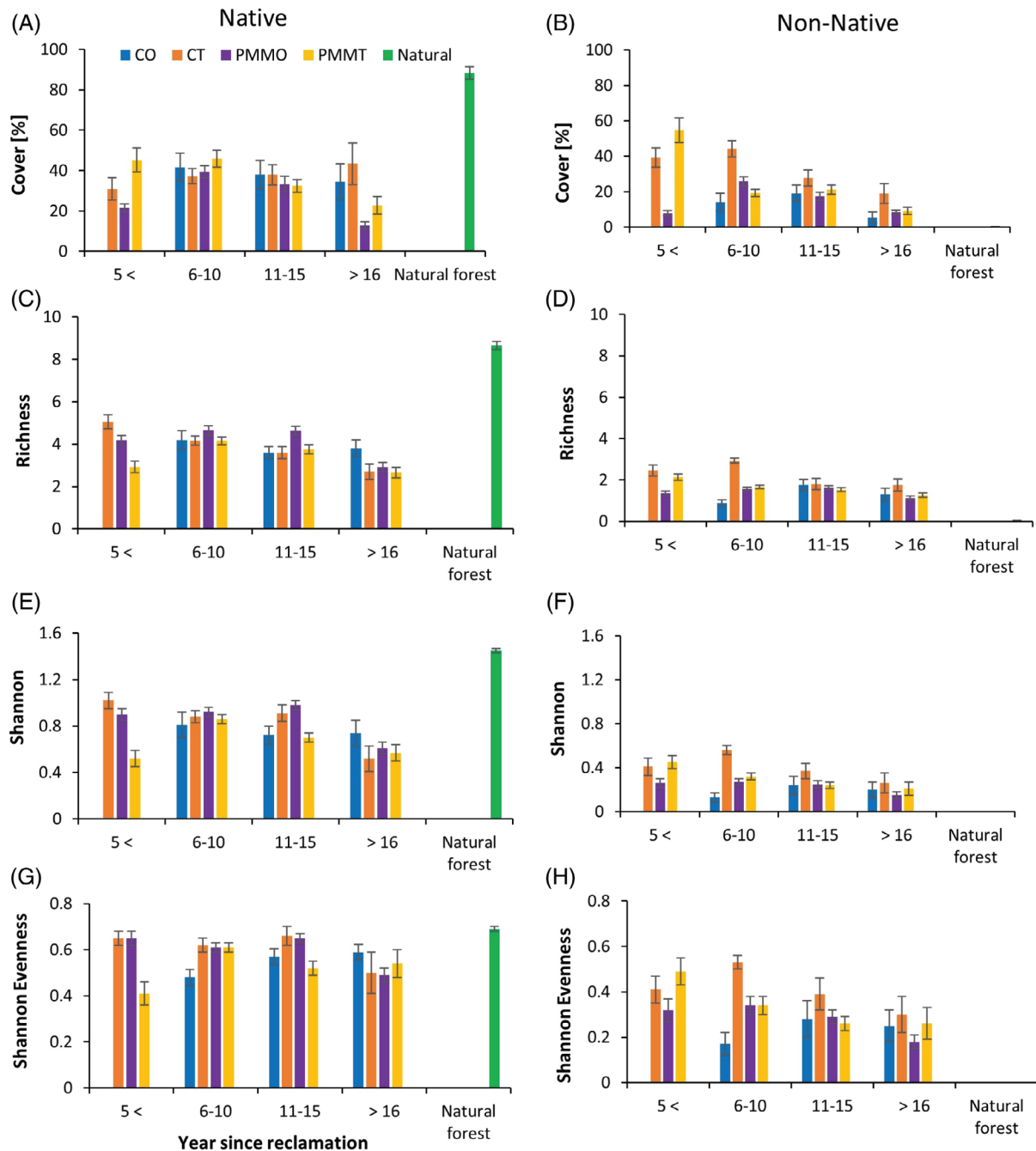


Figure 2. Mean ($\pm$ SE) native and nonnative species percent cover (A, B), species richness (C, D), and Shannon (E, F) and Shannon evenness (G, H) by soil treatments relative to natural forest and age. CO, clay over overburden; CT, clay over tailings sand; Natural, natural forest; PMMO, peat-mineral mix over overburden; PMMT, peat-mineral mix over tailings sand.

included 46 shrubs (34 in reclamation sites and 29 in natural), 121 forbs (94 in reclamation of which 28 are nonnative and 61 in natural of which four were nonnative), 52 graminoids (48 in reclamation and 12 in natural), and a total 31% of species overlaps between reclaimed and natural forests (Table S2). In total, 181 native (138 in reclamation and 97 in natural) and 39 nonnative (39 in reclamation and five in natural) species were identified in the study areas.

Plant community composition differed significantly between reclaimed and natural forest sites (MRPP: $A = 0.2203$, $p = 0.001$). Further pairwise comparisons showed that natural forest plant community composition differed significantly from reclamation treatments. This is also evident in the NMDS ordination where natural forest separated clearly from the reclamation treatments (Fig. 4). This ordination yielded a two-dimensional solution and explained 76.8% of total variation

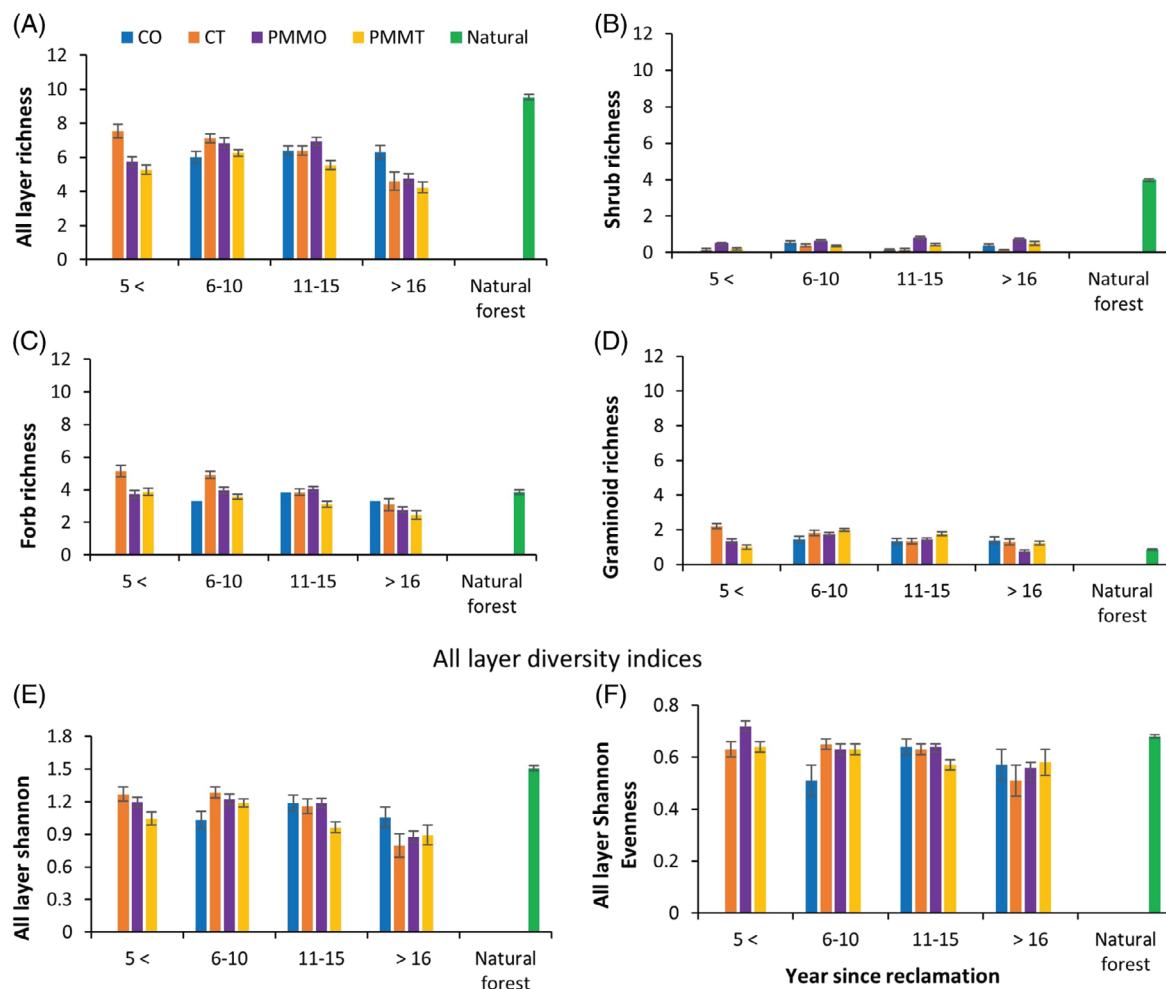


Figure 3. Mean ($\pm$ SE) species richness for all layers (A), shrub (B), forb (C), graminoid (D), and Shannon (E) and Shannon evenness (F) for all layers by soil treatments relative to natural forest and age. CO, clay over overburden; CT, clay over tailings sand; Natural, natural forest; PMMO, peat-mineral mix over overburden; PMMT, peat-mineral mix over tailings sand.

in the original data matrix and revealed some compositional changes over time with no consistent trends.

PCA equilibrium circle analysis showed that reclamation treatments of peat-mineral mix with tailings or overburden had fewer species that significantly contributed to the ordination than natural forest sites. Most were early successional forbs or graminoids, while in the natural forest most species were perennial shrubs or forbs (Fig. S4). In total, four species in PMMT (*Taraxacum officinale* Weber, *Sonchus* sp., *Lotus corniculatus* L., *Poa palustris* L.), seven species in PMMO (*T. officinale*, *P. palustris*, *Sonchus* sp., *L. corniculatus*, *Chamaenerion angustifolium* (L.) Scop., *Rubus idaeus* L., *Calamagrostis canadensis* (Michx.) Beauv.), and 10 species in natural forest showed significant contributions in the ordination and had vectors outside of the equilibrium circle.

Pattern of Species Co-occurrence

The probabilistic model of species co-occurrence analyzed 20,699 (PMMO 8,911, PMMT 5,460, natural 6,328) species

pairs of which 5.4, 4.4, and 24.9%, respectively, showed significant association (positive or negative). Less than 1% showed significant association in clay tailings and clay overburden reclamation treatments (possibly due to smaller sample size; Fig. S5). The potential pairwise association was slightly higher than expected (5%) in PMMO and natural forest, and lower than expected in PMMT. Although the natural forest had a higher number of negative and positive associations, in all instances the total number of positive associations was higher than the negative associations. Analysis of species co-occurrence patterns based on pairing of major tree species with understory revealed that *Populus tremuloides* and *Picea glauca* had only positive associations; *Populus balsamifera* had both positive and negative associations; and *Pinus contorta* had only negative associations with understory species in both peat-mineral treatments (Table S3). Interestingly, no positive or negative associations were documented for *Pinus banksiana*. In PMMO *P. tremuloides* and *P. balsamifera* and in PMMT *P. glauca* and *Pinus contorta* were the major tree species that showed significant

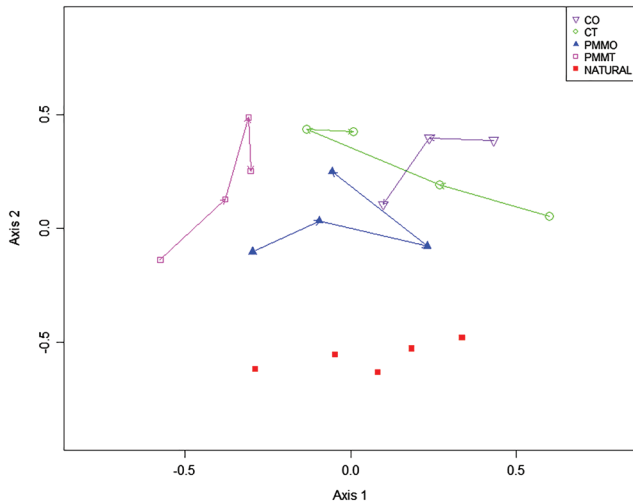


Figure 4. Nonmetric multidimensional scaling ordination showing plant community composition change for year since reclamation group by soil treatments relative to natural forest. CO, clay over overburden; CT, clay over tailings sand; Natural, natural forest; PMMO, peat-mineral mix over overburden; PMMT, peat-mineral mix over tailings sand.

associations with the understory. Forbs and graminoids were associated with *P. balsamifera* and *P. glauca* while shrubs and forbs were associated with *P. tremuloides* (Table S3). In the natural forest, *Betula papyrifera*, *P. banksiana*, *P. balsamifera*, and *P. tremuloides* had both positive and negative associations and *Abies balsamea* had only negative associations with the understory (Table S3). The greatest number of significant associations was observed for *P. banksiana* (eight positive and 21 negative) followed by *P. tremuloides* (12 positive and three negative) and *P. balsamifera* (seven positive and eight negative). Most of the associations were with either shrubs or forbs in the natural forest. Species associations in the natural forest reflect the types of sites that specific species are associated with; e.g. tree species *P. banksiana* had positive associations with dry site species and negative associations with mesic site species (Table S3).

Discussion

Although there was variation in measurable attributes of plant communities for the reclamation treatments and natural forest, most of the reclamation treatments followed the same general trend in ecosystem recovery as natural boreal forests in this region (Hart & Chen 2006). Differences between reclaimed sites and the natural forest are likely due to effects of soil moisture regime, soil properties, subsoil characteristics, types of cover soil used for reclamation, depth of cover, plant propagules in the seed bank, soil interspecific interactions, and time (Azeria et al. 2011; Pinno & Hawkes 2015; Dhar et al. 2018). Cover soil examined in this study was either clay or peat-mineral mix which may have a direct impact on the availability of propagules. Several studies have reported that peat-mineral mix cover soil has fewer propagules than the natural forest (MacKenzie et al. 2012; Macdonald et al. 2015a; Errington &

Pinno 2016). For example, MacKenzie and Naeth (2010) found that peat-mineral soil (approximately 3,600 seeds/m²) contained less seeds than natural forest floor soil (approximately 9,000 seeds/m²). There is currently no information regarding plant propagules present in the seed bank of clay cover soil.

In boreal forest ecosystems, newly disturbed areas tend to be dominated by forbs and graminoids with shrubs becoming a larger component of the ecosystem with time after reclamation (Rowland et al. 2009; Pinno & Hawkes 2015). This was evident in our study as cover and richness of forb and graminoid layers in most reclamation treatments were similar to natural forest within 16–24 years after reclamation relative to shrubs. This probably results from reduced resource availability in the understory with increasing overstory canopy closure and standing biomass (Audet et al. 2015; Pinno & Hawkes 2015; Zhang et al. 2017). A similar observation reported by Corns and La Roi (1976) concluded that species diversity and richness generally increase initially following stand replacing disturbance, then decrease as the forest canopy closes. In contrast, lower shrub cover and richness are primarily due to slow establishment and growth of shrubs on reclaimed sites, rather than regeneration from bud banks and seed banks following natural disturbances (Hart & Chen 2008; Pinno & Hawkes 2015). Reductions in shrub cover and richness may also be due to invasion of shade intolerant and ruderal species that established quickly after cover soil placement. Such species may not persist in understories, but germinate from the seed bank or disperse onto vacant seed beds after disturbance due to differences in resource availability (light, substrate; Hart & Chen 2008; Craig & Macdonald 2009). Although some common boreal shrubs (e.g. *Cornus canadensis* L.; *Linnaea borealis* L.; *Viburnum edule* (Michx.) Raf.; and *Alnus crispa* (Ait.) Pursh) are missing in the reclamation treatments, the progression of shrub cover and richness toward natural forest conditions suggests that recovery of shrubs is still progressing and may simply require additional time, similar to what has been reported in other studies (Rowland et al. 2009; Pinno & Hawkes 2015; Dhar et al. 2018, 2019). The presence and cover of shrubs may be a good indicator of reclamation success as they accelerate forest floor development, facilitate tree establishment through snow retention, reduce drought, temperature stress, and seedling mortality from browsing (Rowland et al. 2009). The nitrogen fixing shrubs can increase soil nitrogen pools and soil organic matter (Mummey et al. 2002) which could improve overall soil chemical and biological properties in reclamation sites (Rowland et al. 2009). Establishment of nitrogen fixing shrubs and other species has been recommended to improve chemical and biological properties of reclaimed soils (Lefrancois et al. 2010). A nitrogen fixing native shrub *Shepherdia canadensis* (L.) Nutt. was found in all reclamation treatments and was positively associated with *Populus tremuloides* in PMMO treatment.

The functional groups to which the species belong are an important aspect of characterizing plant communities as they potentially affect the successional trajectory of reclamation sites (Hart & Chen 2006; Dhar et al. 2018, 2019). Generally, tree and shrub canopies shade out ruderal species (Liefers et al. 1999) and can reduce competition and promote development of

forest understory species. In our study, reclamation treatments showed an affinity for perennial forbs (*Taraxacum officinale*, *Sonchus* sp., *Lotus corniculatus*, *Chamaenerion angustifolium* (L.) Scop.), shrubs (*Rubus idaeus*), and grasses (*Poa palustris*, *Calamagrostis canadensis*). Thus, the trajectory of community development on reclamation treatments appears to follow the typical early successional pattern or recovery process of boreal forests from ruderal and annual to perennial communities (Liefers et al. 1999; Hart & Chen 2006). The presence of indicator native shrub species *Rubus idaeus* in all reclamation treatments also suggests that the successional trajectory after reclamation follows the typical boreal forest recovery as this shrub is often found in recently disturbed boreal forest communities (Hart & Chen 2006) and may be a good indicator of reclamation success (Chen et al. 2017).

In our study, nonnative species cover declined over time (34% at age group <5 years to 10% at >15 years) which is comparable with the findings of Pinno and Hawkes (2015) but higher than (5%) Dhar et al. (2019) found 20 years after oil sands reclamation. The declining trend indicates that nonnative species might be exploiting the resources available after land reclamation but may be eliminated over time as native species develop. Nonnative species are often problematic in disturbed ecosystems as they may alter ecosystem development by influencing community composition, biodiversity (Wilson & Pinno 2013), and ecosystem processes such as nutrient cycling (MacDougall & Wilson 2011), although their early establishment does provide cover, help retain soil, and contribute to the development of soil structure and an accumulation of soil organic matter. Therefore, ongoing monitoring is recommended to focus on the potential long-term impacts of nonnative species on the ecosystem recovery of reclaimed sites and their influences on plant community development.

Most shrubs and forbs found in the reclamation treatments are the preferred species to use for reclamation in Alberta (Smreciu et al. 2013). However, nonnative *Melilotus* and a few grass species were also positively associated with *Populus balsamifera* and *Picea glauca* in both peat-mineral mix reclamation treatments and are of concern due to potential influences on development of other understory vegetation, and active management may be required to control these species. In contrast, *P. tremuloides* was positively associated with nitrogen fixing (species *S. canadensis*) or erosion controlling species (*Salix bebbiana*) which could contribute to further increases in diversity and establishment of other important or priority species by moderating abiotic conditions (e.g. exposure to extreme temperatures, improved soil drainage and water holding capacity, suppression of grass) and influencing soil conditions, through faster decomposition and increased soil carbon accumulation (Dhar et al. 2018; Jean et al. 2019).

While many studies report that plant community assembly after natural disturbance is nonrandom (Drake 1990; Azeria et al. 2011), our study shows that in contrast to natural forested ecosystems, reclaimed ecosystems had a higher random pattern of species association. A similar pattern of species association was observed in several other studies following open cast lignite mine reclamation in northeastern Germany (Zaplata et al.

2013; Ulrich et al. 2016) and a glacial outwash plain in Iceland (Marteinsdóttir et al. 2018). This suggests random aggregations of species are able to tolerate the same environmental conditions (Gleason 1926) and thus existing species are more likely to allow new (invading) species that are functionally similar to the existing species.

A higher proportion of nonrandom patterns of species association in the natural forest suggests competitive exclusion of species where establishment of certain species were inhibited due to their having resource requirements similar to the existing species (Tilman & Pacala 1993; Rees et al. 2001; Dhar et al. 2017). Assembly rules and successional dynamics of natural forest conditions may be different from those of reclaimed sites in this part of the boreal forest, due to disturbance intensity and its influences on overall biological legacies, which play a significant role in postdisturbance plant community development (Drever et al. 2006; Dhar et al. 2016). After natural disturbances such as fire, biological legacies (surviving trees, snags, logs, patches of intact vegetation, seedbanks in tree crown or soil) generally allow the forest to retain its genetic, compositional, and structural predisturbance conditions (Drever et al. 2006; Dhar et al. 2016). Thus, communities that developed following natural disturbances such as fire are more structured and strongly inhibit invasion of new species that are functionally similar to the resident species (Fargione et al. 2004). This is not the case for oil sands reclamation sites as mining and subsequent reclamation severely alter soil properties and biological legacies and thus plant community assembly in oil sands reclamation areas in Alberta is different and remains largely unstructured 16–24 years after reclamation. Although continued monitoring is required to quantify ecosystem recovery following reclamation of oil sand mine, the outcomes from this study can be used to prioritize future research and operational activities for oil sands reclamation studies.

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Supporting Information

The following information may be found in the online version of this article:

Table S1. Results of mixed effects models for cover, diversity, and richness of vegetation layer.

Table S2. List of species found in the reclamation and natural forest area.

Table S3. Significant pairwise associations between major trees and other species.

Figure S1. The LTPN plot distribution in the Fort McMurray oil sands region.

Figure S2. Long Term Plot Network (LTPN) permanent plot (10 × 40 m) layout used for data collection.

Figure S3. Spaghetti plots showing the temporal trends in plant community cover change for mesic PMMT ($n = 6$) and PMMO ($n = 11$) treatment plots that have repeated measurements.

Figure S4. Equilibrium circles for the first two PCA axes for peat-mineral mix tailings, peat-mineral mix overburden, and natural forest.

Figure S5. Expected and observed co-occurrence of peat-mineral mix tailings, peat-mineral mix overburden, and natural forest.

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